Chapter 2



PLC Hardware Components

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Objectives

- MLO #2-11: Describe the circuitry for discrete input and output modules of A-B and Click PLC.
- MLO #2-12:Understand analog input and output modules.
- MLO #2-13: Name the PLC basic general instructions
- MLO #2-14: Name the factors considered in the selection of a PLC system
- MLO #2-15: Indicate the differences between A-B PLC and A-B PAC.
- MLO #2-16: Name and describe the common industrial automation systems. Establish when to use them.

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Section 2.2

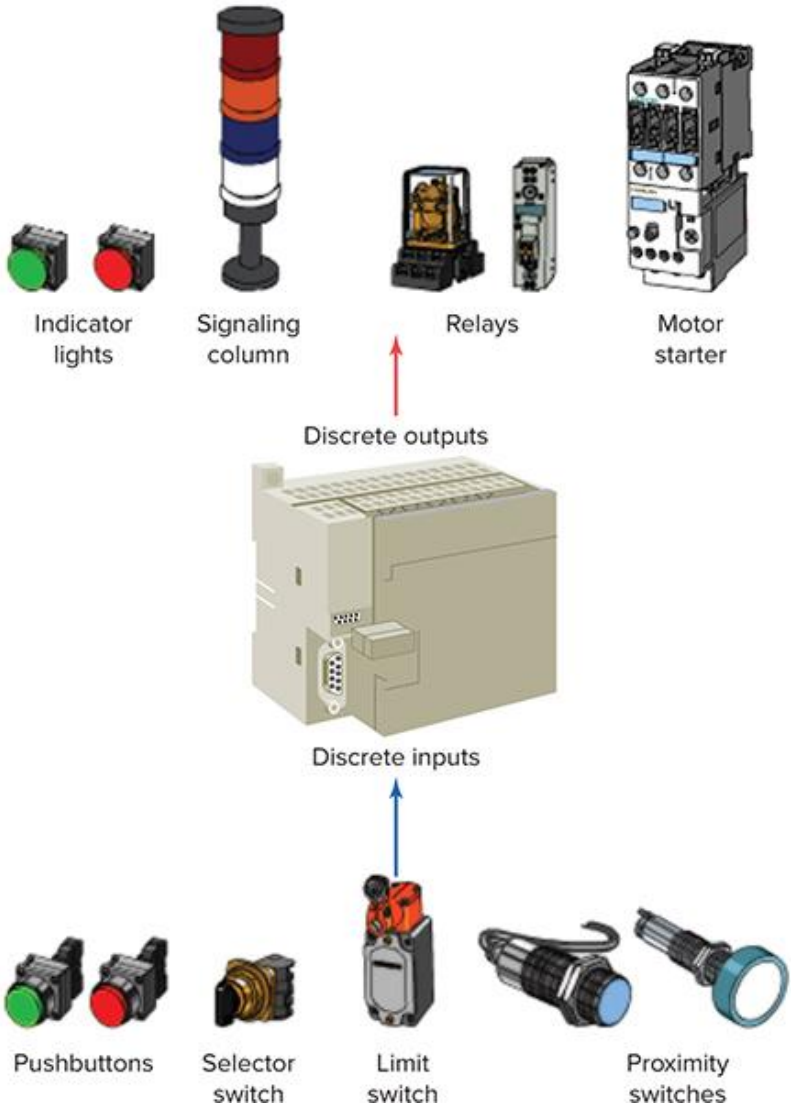
Discrete I/O Modules

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Discrete field devices

The *discrete I/O* interface module connects field input devices of the *ON/OFF* nature.

The classification of discrete I/O covers *bit oriented* inputs and outputs.



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A-B Discrete I/O voltages

Each discrete I/O module is powered by some *field supplied* voltage source.

Since these voltages can be of different magnitude or type, I/O modules are available at various AC and DC voltage ratings.

Table 2-1 Common Ratings for Discrete I/O Interface Modules

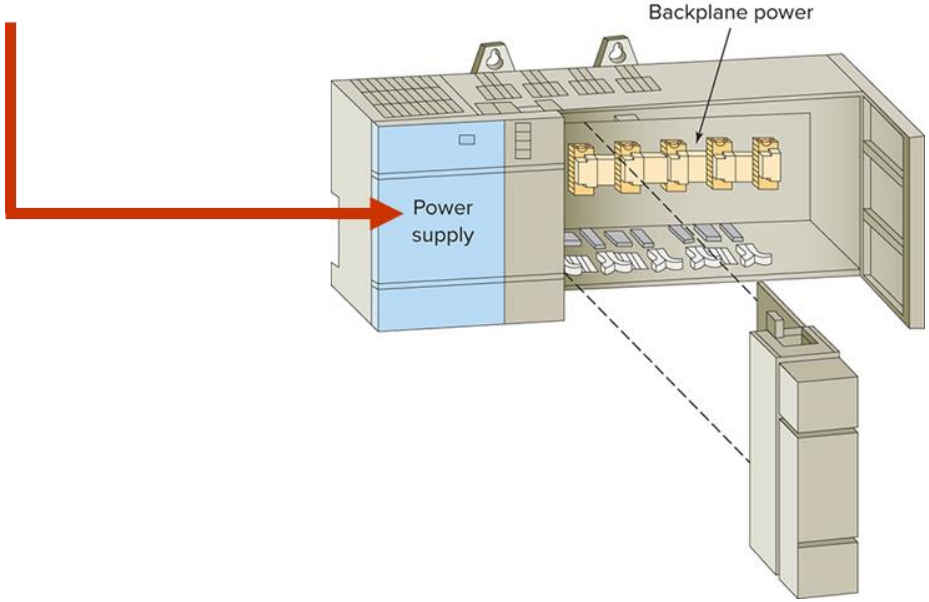
Input Interfaces	Output Interfaces
12 V **AC/DC** /24 V **AC/DC**	12-48 V **AC**
48 V **AC/DC**	120 V **AC**
120 V **AC/DC**	230 V **AC**
230 V **AC/DC**	120 V **DC**
5 V **DC** (TTL level)	230 V **DC**
	5 V **DC** (TTL level)
	24 V **DC**

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Backplane voltage

Modules receive their voltage and current operation from the backplane of the rack enclosure into which they are inserted.

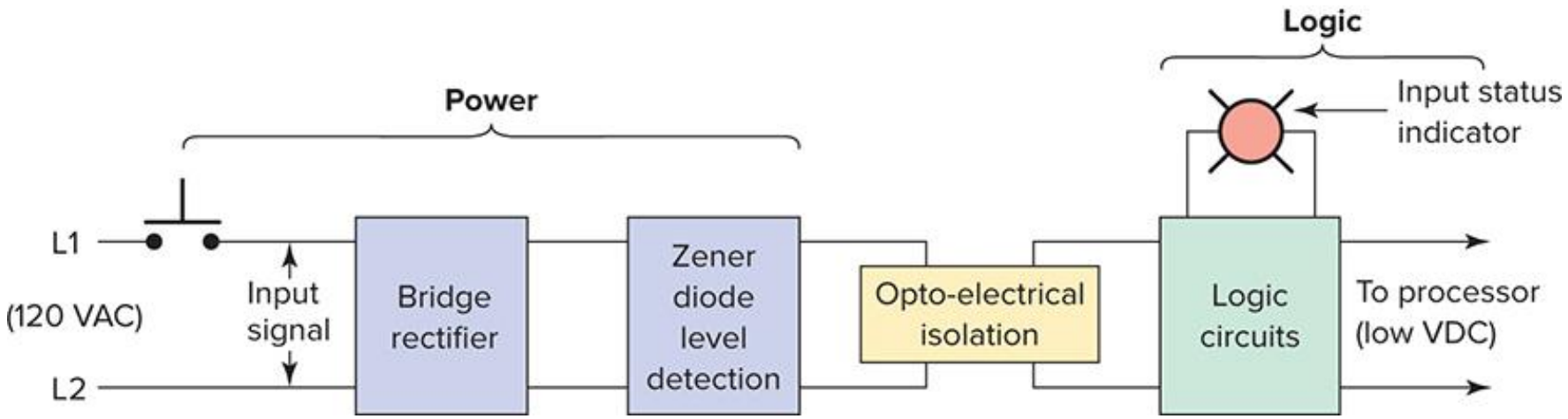
Backplane power is provided by the PLC module power supply and is used to power the electronics that reside on the I/O module circuit board.



Currents required by the loads are normally provided by user-supplied power.

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A-B AC *discrete input module* block diagram.

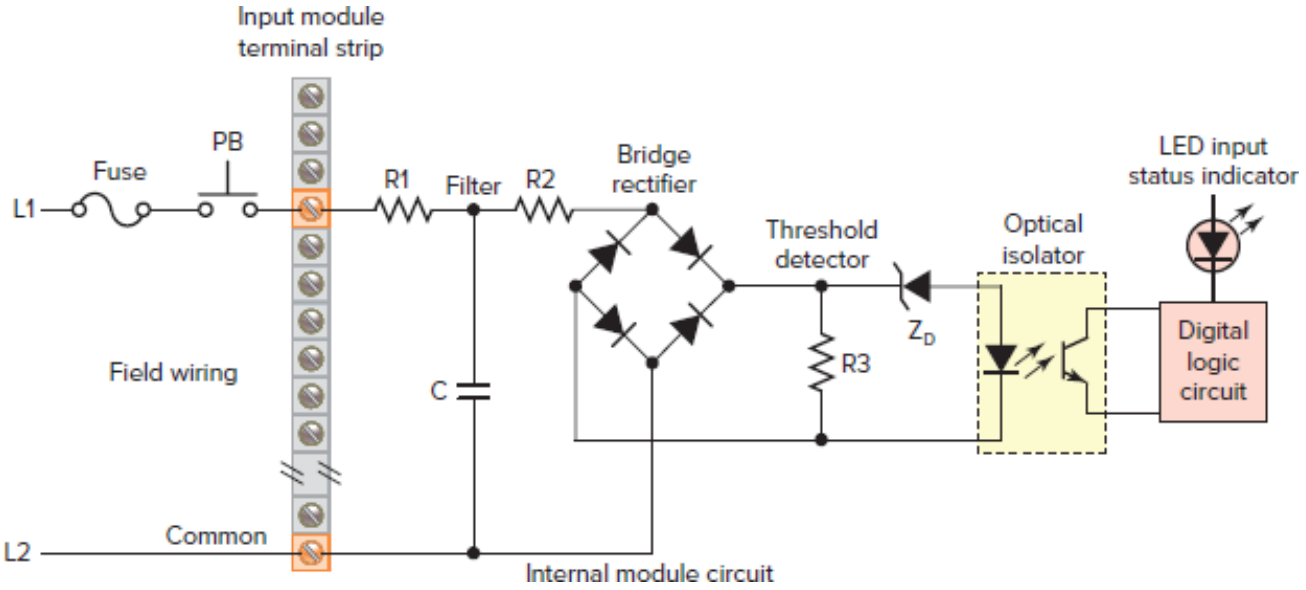


- **The circuit is made up of a *power* section and a *logic* section.**
- **An *optical isolator* is used to provide electrical isolation between the power and logic circuitry.**
- **The input LED turns on or off, indicating the *status* of the input device.**

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AC *discrete input module* schematic diagram.

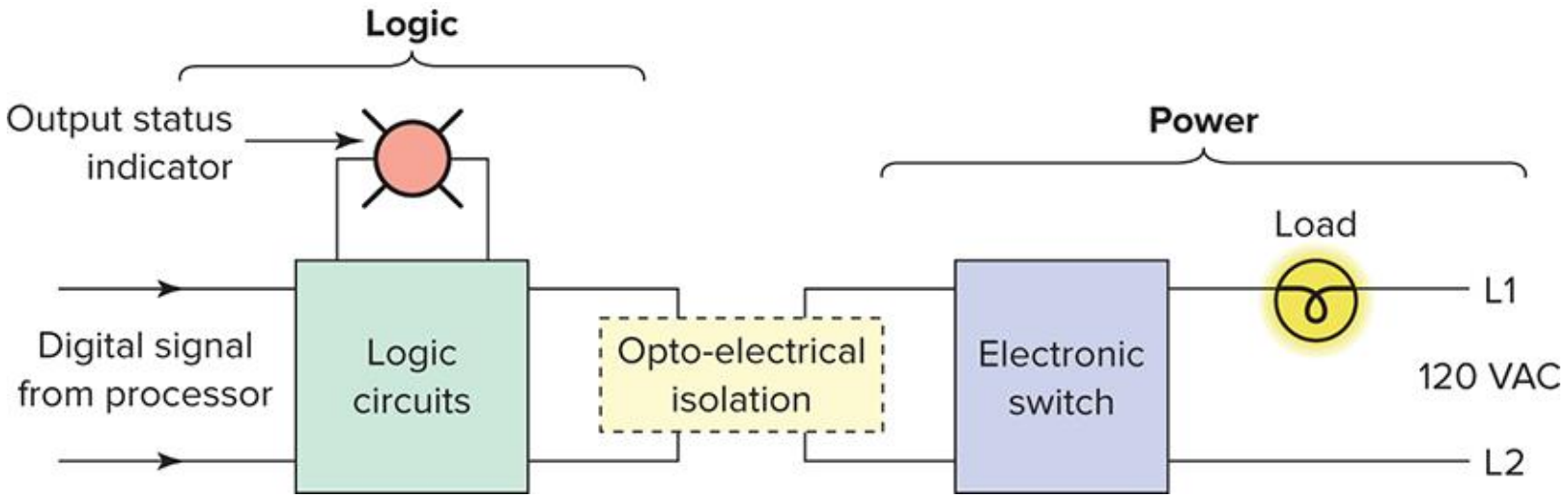


- **When the pushbutton is closed AC is applied to the bridge rectifier input.**
- **This results in a low-level DC output voltage that is applied across the LED of the optical isolator.**
- **When light from the LED strikes the phototransistor, it switches it into conduction.**

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AC *discrete output module* block diagram.

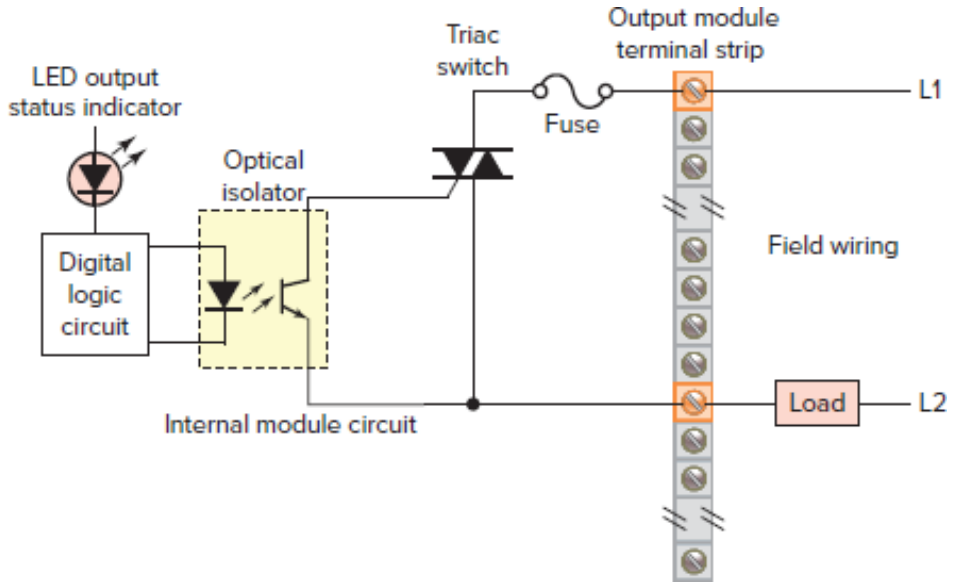


- **The module is composed of the power section and the logic section, coupled by an isolation circuit.**
- **The power output interface can be thought of as an electronic switch that turns the output load device on and off.**

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AC *discrete output module* schematic diagram.



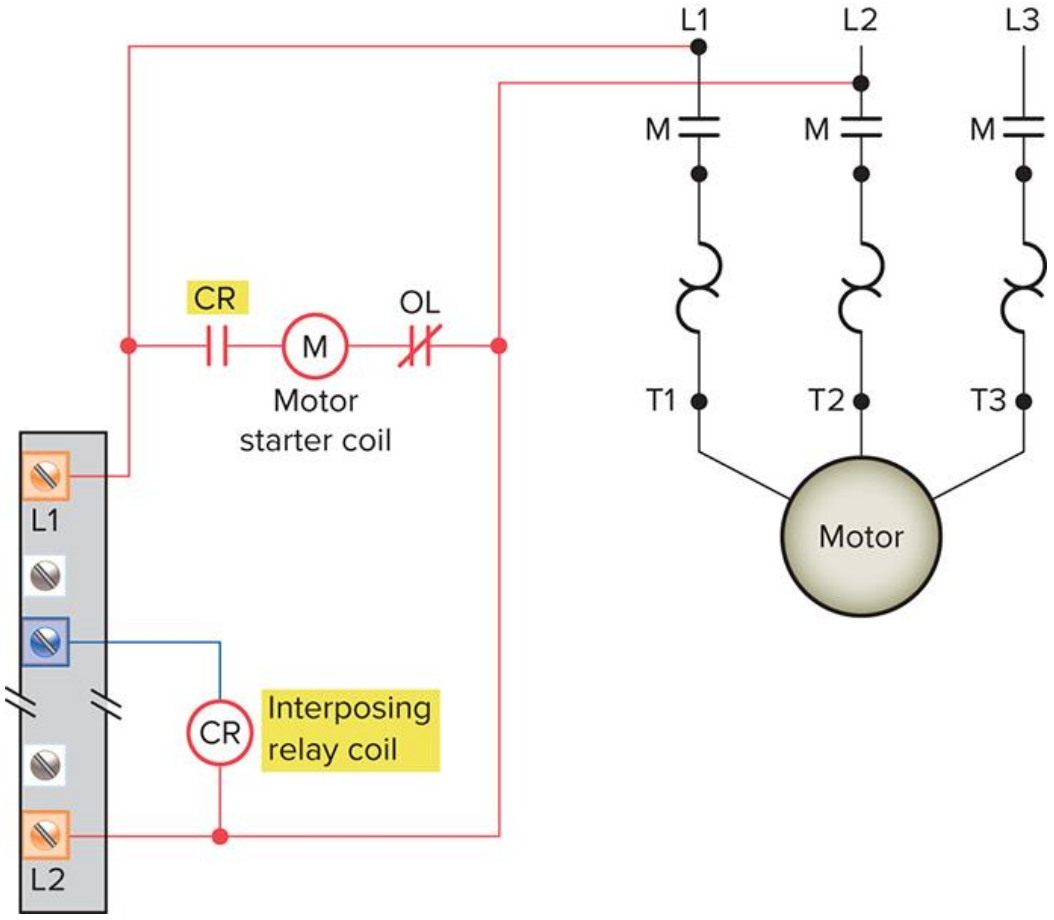
- **When the processor calls for the output load to be energized, a voltage is applied across the LED of the opto-isolator.**
- **This in turn triggers the triac AC semiconductor switch into conduction allowing current to flow to the output load.**

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Individual AC outputs are usually limited by the size of the triac to *1 A or 2 A*.

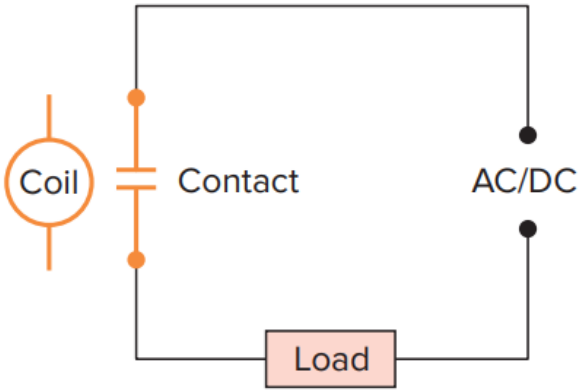
For controlling larger loads an *interposing relay* is connected to the output module. The contacts of the relay can then be used to control a larger load or motor starter,



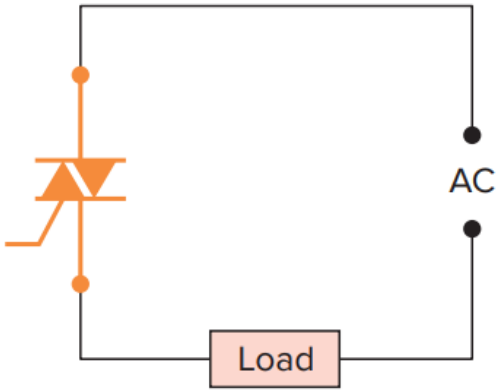
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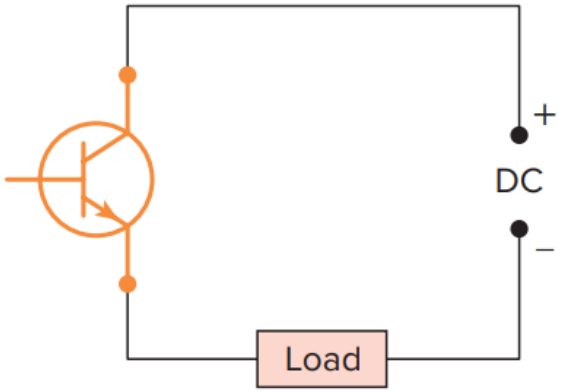
Discrete output modules are used to turn field output devices either *on or off*.



Relay output



Triac output



Transistor output

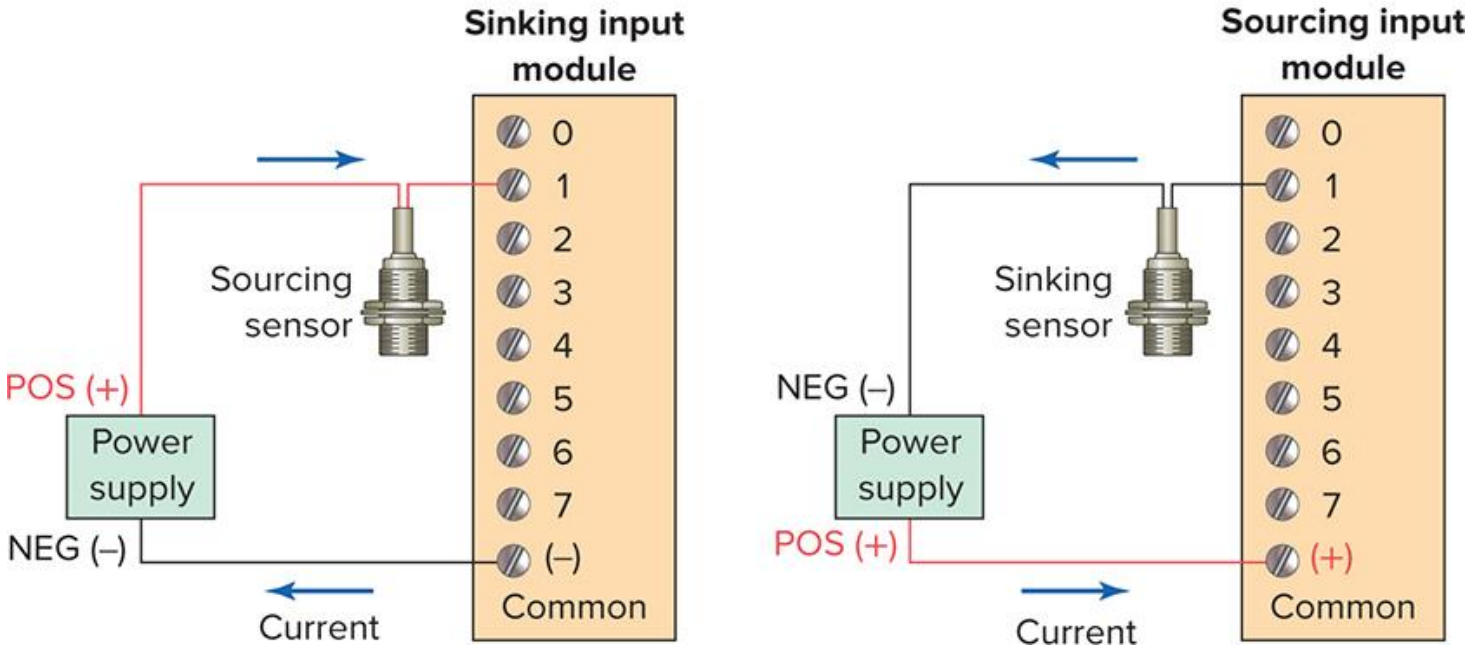
Output modules can be purchased with *transistor*, *triac*, or *relay* output

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Current source or sinking input modules

Certain DC I/O modules specify whether the module is designed for interfacing with *current-source* or *current-sink* devices.



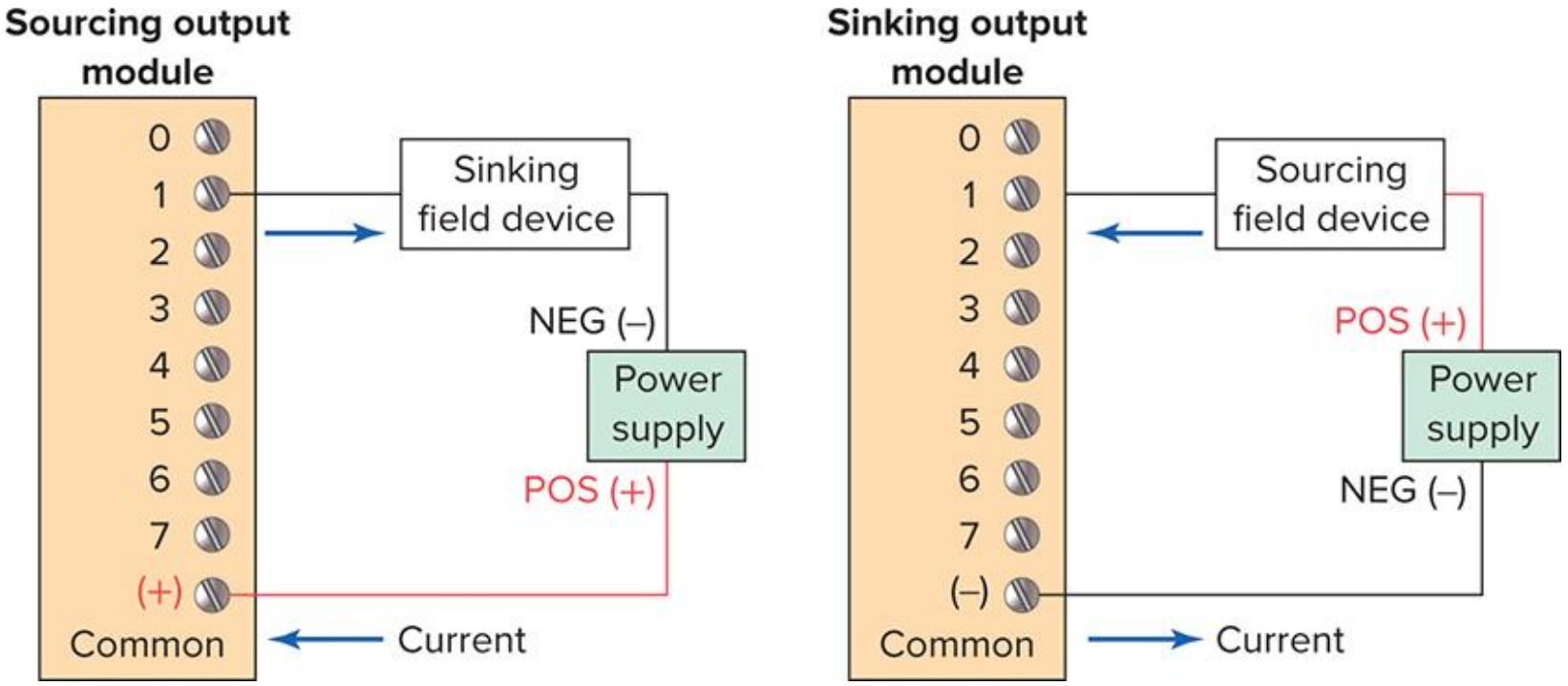
If the module is a *current-sourcing module*, then the input or output device must be a *current-sinking device*.

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Current sinking – current sourcing

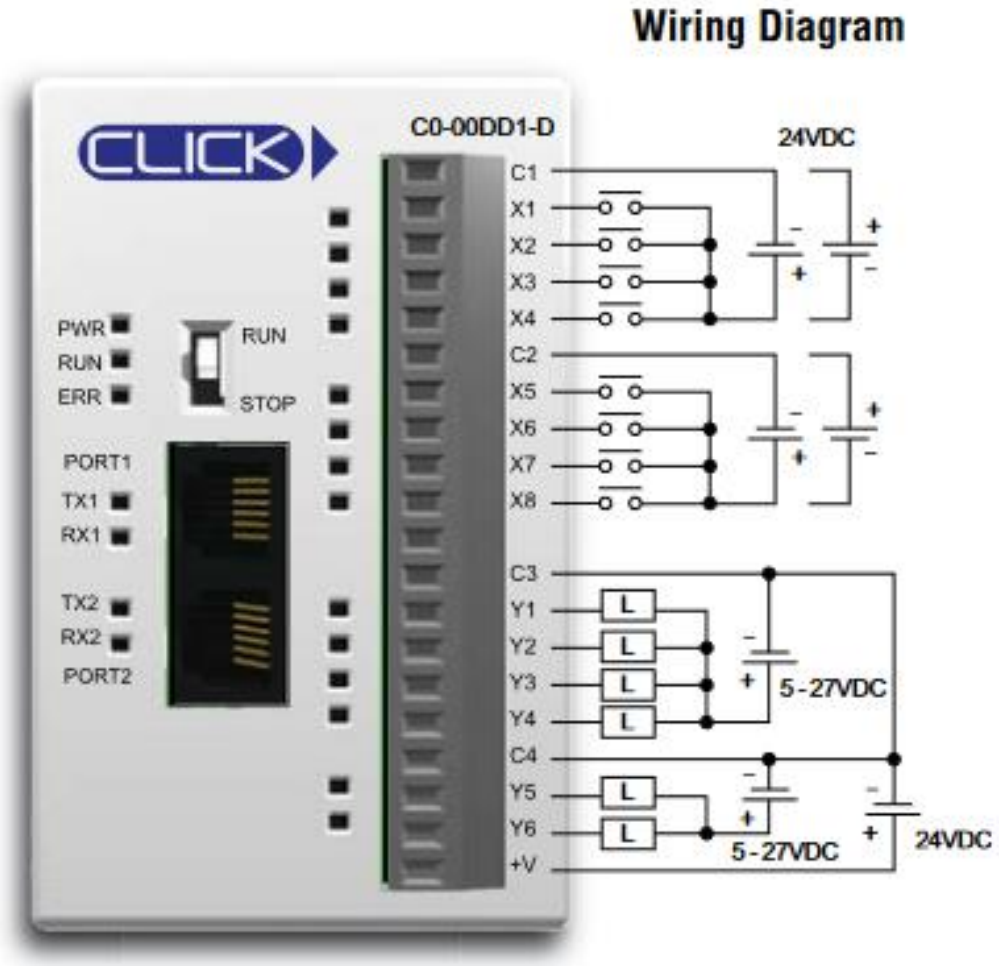
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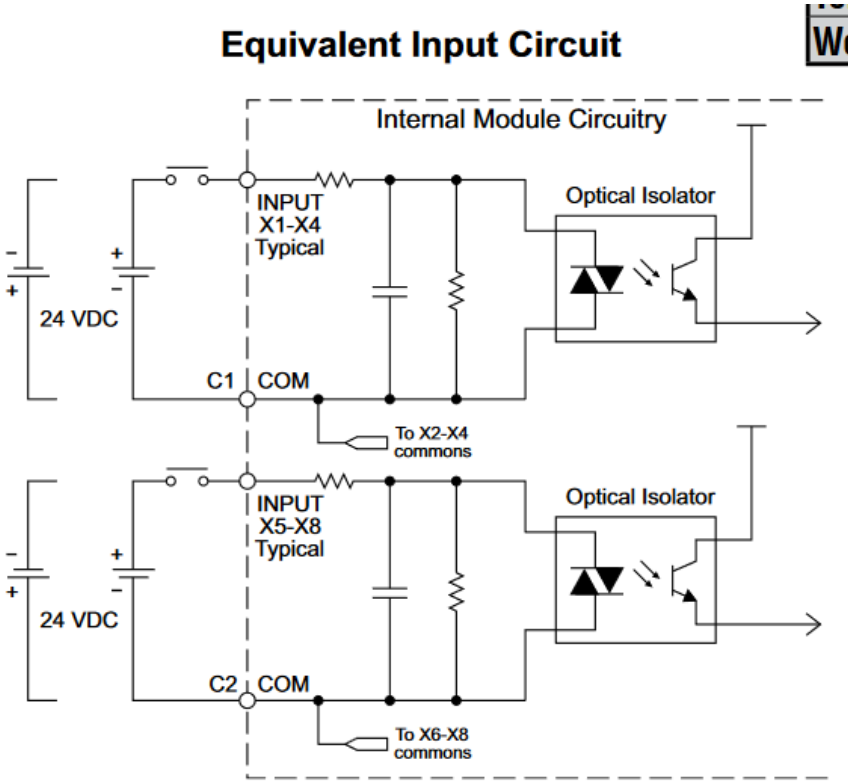
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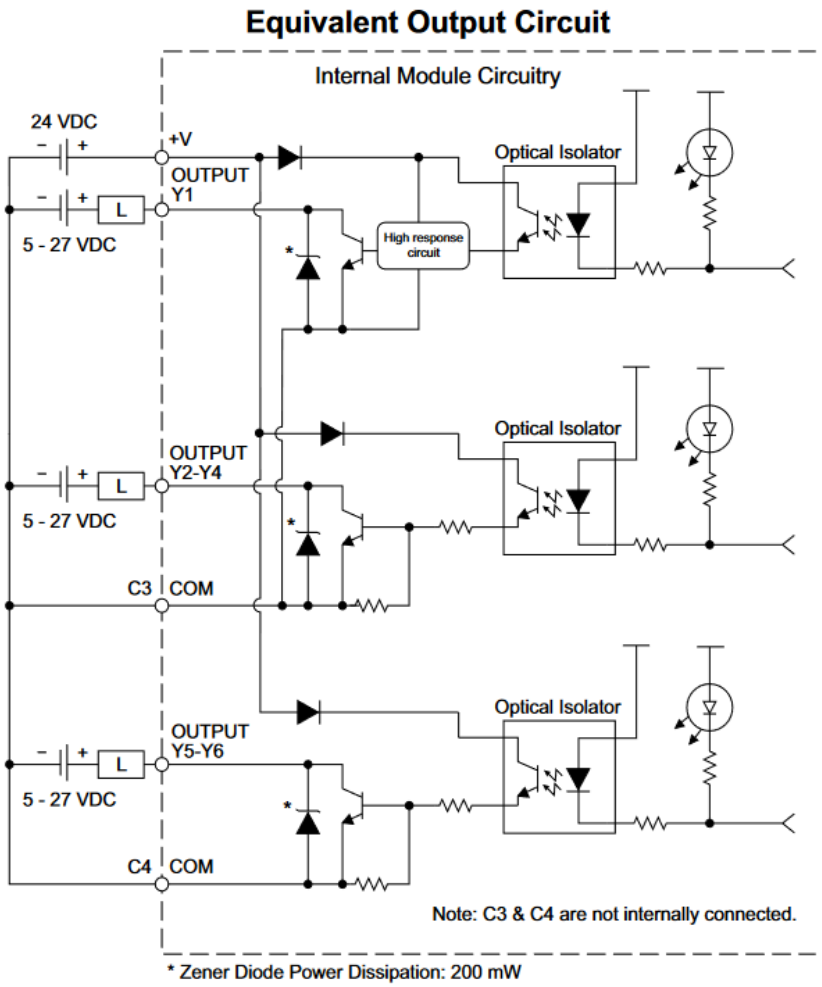
Click C0-00DD1-D Discrete Modules



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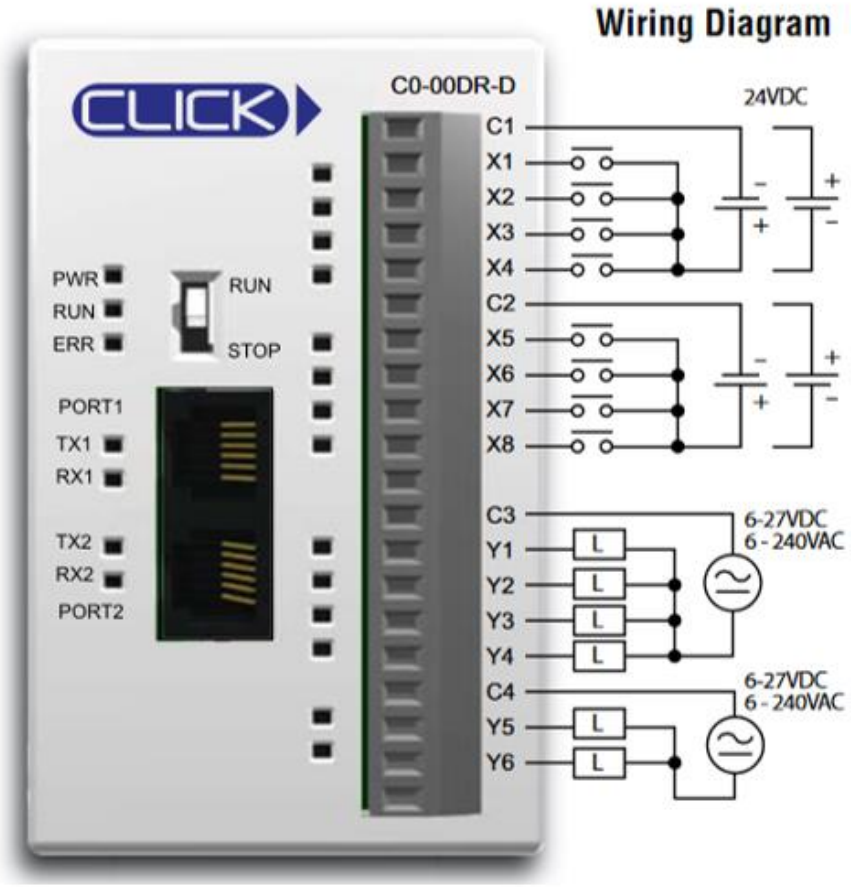
Click C0-00DD1-D Discrete Modules





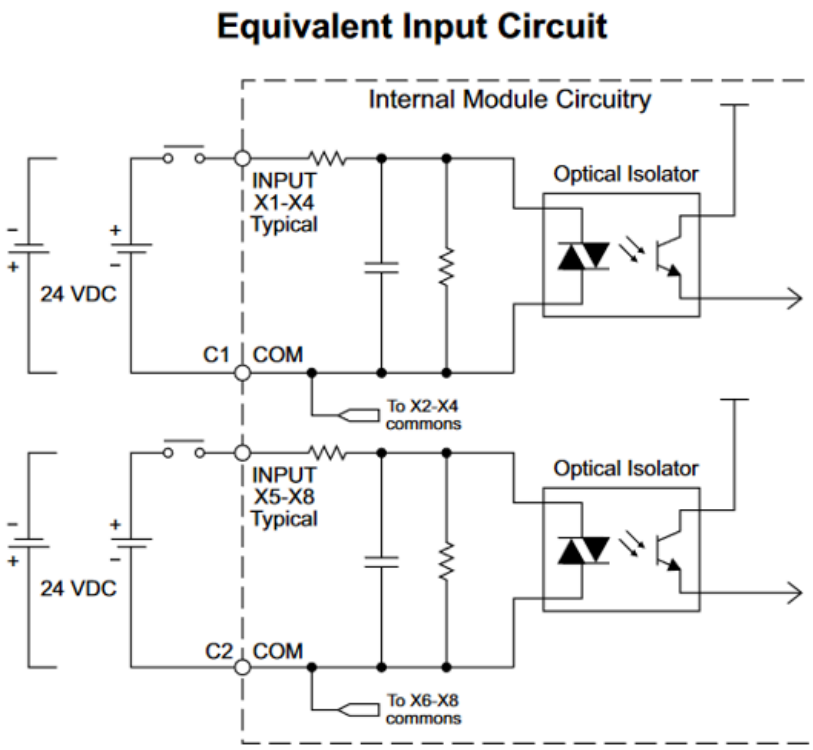
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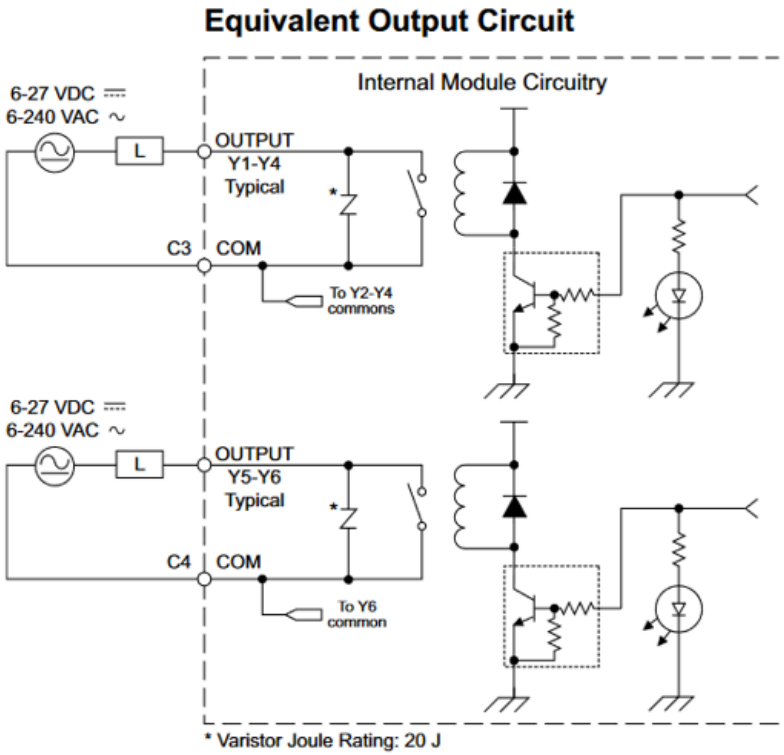
Click C0-00DR-D Discrete Modules



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Click C0-00DR-D Discrete Modules





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Analog Input/Output Modules

- Analog-to-digital converter (ADC):
  - Converts analog input signals to digital signals
- Digital-to-analog converter (DAC):
  - Converts digital output signals back to analog signals
- This class will focus only on discrete I/O signals

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PLC General Instructions

- Programmable logic controllers (PLCs) are used to implement specific instructions:
  - Input/output (I/O) port control
  - Timing
  - Counting
  - Data manipulation
  - Reporting
  - Arithmetic
  - Logic
  - Communication

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A-B Instruction set

The *instruction set* for a particular PLC lists the different types of instructions supported.

Symbol	Name	Bit status	Instruction status
	XIC	0	FALSE
		1	TRUE
	XIO	0	TRUE
		1	FALSE

Table 1-1 Typical PLC Instructions

Instruction	Operation
XIC (Examine ON)	Examine a bit for an ON or 1 condition
XIO (Examine OFF)	Examine a bit for an OFF or 0 condition
OTE (Output Energize)	Turn ON a bit (nonretentive)
OTL (Output Latch)	Latch a bit (retentive)
OTU (Output Unlatch)	Unlatch a bit (retentive)
TOF (Timer Off-Delay)	Turn an output ON or OFF after its rung has been OFF for a preset time interval
TON (Timer On-Delay)	Turn an output ON or OFF after its rung has been ON for a preset time interval
CTD (Count Down)	Use a software counter to count down from a specified value
CTU (Count Up)	Use a software counter to count up to a specified value

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PLC and Industrial Automation

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Factors to Consider in the Selection of a PLC System

- Depending on the application, the engineer must select a PLC based on:
  - Number of Inputs/outputs (I/O size)
  - Processor and speed
  - Memory size
  - Maintenance and support
  - Cost

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A-B Programmable Automation Controller (PAC)

- Programmable automation controller (PAC):
  - Personal computer (PC) with PLC architecture
  - Can be used to perform PLC instructions
  - Control electric drives and more sophisticated motion and batch control systems
  - More advanced programming features.
  - Programming structure is very different

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Common Industrial Automation System

- PLC systems are traditionally used for discrete, logic-based control and automation using ladder logic programming. Generally, PLCs are used in machine tools, operation sequencing, and manufacturing and process control.
- Programmable Automation Controllers (PACs) are similar to high end PLC systems. Essentially, PAC systems include more power, and the programming goes beyond ladder logic. They offer more memory and I/O ports.

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Common Industrial Automation Systems (cont.)

- Distributed Control Systems (DCSs) uses many different controllers installed in many locations. They are all connected to a network and all information is centralized in a dedicated control room. A bulk of analog control is expected.
- Supervisory Control and Data acquisition systems (SCADAs) has distributed remote terminal units (RTUs) that collect information from the remote locations and transmit it a central computer that supervises the state of all the processes monitored by the RTU.

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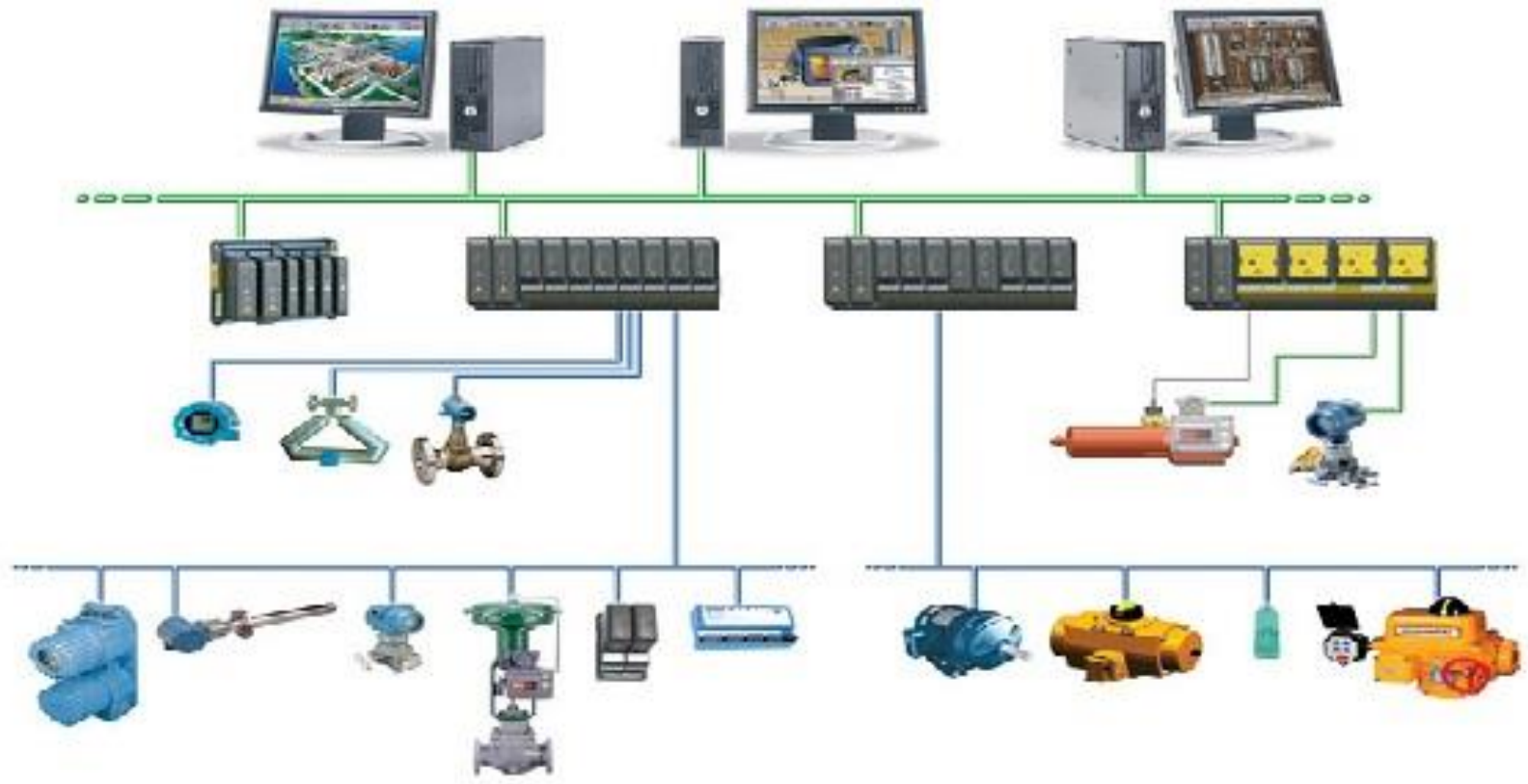
Common Industrial Automation Systems (cont.)

Control System Comparison				
Feature	**PLC**	**PAC**	**DCS**	**RTU**
Industrially hardened hardware	X	X	X	X
Discrete control/sequential control	X	X		X
Process control (batch and continuous)	X	X	X	
Remote monitoring and control, supervisory control and data acquisition (SCADA)	X	X		X
Capable of data-intensive applications	X	X	X	
Floating-point math	X	X	X	
Compact, high-density system	X	X		
Open languages and standard protocols	X	X		
Motion control		X		
Critical local operations continue in the event of a communication failure		X	X	
Multifunction		X		
Multitasking		X		
Efficient processing and I/O scanning		X	X	
Flexible, modular architecture		X		
Extensive communication options		X		X
Open standard network interfaces		X		
Integrated hardware and software		X	X	
Plain-language tag names		X		

Adapted from Opto 22

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DCS, SCADA, PLC/PAC Systems



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DCS, SCADA, PLC/PAC Systems

- Any industry that requires the control or monitoring of many variables (thousands are not uncommon), and the processes are distributed in a large area.
- Data acquisition and control is distributed but supervision is centralized. A local network makes sure data flows among the components of the system.
- As an automation engineer you will wire and connect the instruments to the local control units, configure the system, tune the controllers, set the human-machine interface (HMI), train operators, etc. Programming is at a high level of abstraction.

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Common Industrial Automation Systems (cont.)

- At the end of the day, all these systems need to communicate through a network
- All relevant information must be transmitted on time for their processing and monitoring.
- Operators must be able to visualize all this information and take measures when needed.
- Expanding is possible by connecting more devices to the network.
- The divisor lines between these automation systems are becoming blurrier.

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